

V.F.S. v2.0 (Open-Gate)

Expanded Core Formulas, Parameters, and Sophianic Folding Feedback

Hybrid core compendium

Hybrid core compendium with Sophianic Folding refinement

Scope

This file collects the **expanded core Open-Gate formulas of V.F.S. v2.0** and preserves the original index-derived structure. The Sophianic Folding layer is integrated as a later Open-Gate refinement rather than as a replacement for the existing core. Formulas that belong specifically to `appendix.html` or `lyapunov.html` remain excluded unless needed to explain the core feedback structure. Graphs, numerical plots, and visual phase diagrams are also excluded.

The governing distinction is:

V.F.S. v2.0 Open-Gate: $\dot{\lambda} = -\dot{\sigma} + I_{\text{gate}}, \quad \sigma + \lambda = C_0 + \mathcal{G}_{\text{recepta}}(t).$

The closed Microcosm law $\dot{\lambda} = -\dot{\sigma}$, $\sigma + \lambda = \text{const}$ is only the limit $\zeta_0 = 0$.

1 Core State Variables

Symbol	Name	Meaning in V.F.S.
$V(t)$	Voluntas	Will; inner orientation and willing.
$F(t)$	Factum	Action; embodied execution of will.
$\sigma(t)$	Singular Resistance	Resistance, deformation, or singular obstruction.
$\lambda(t)$	Sophia	Wisdom; transformed resistance plus received grace in Open-Gate.
$P(t)$	Pleroma	Dynamic fullness of the Vessel.
$S(t)$	Being / Sum / Gignesthai	First derivative of Pleroma; rate of becoming.
$\Omega_P(t)$	Brim	Dynamic capacity / rim of the Vessel.
$H_K(t)$	Gate memory	Acquired receptivity; the learned openness of the gate.
$W(t)$	Wound memory	Vulnerability / wound-memory layer paired with H_K .

$q_{\text{fold}}(t)$	Folding amplitude	Morphological shape-mode of the Vessel; zero in the old form and nonzero in Sophianic Folding.
$Q_{\text{fold}}(t)$	Folding intensity	Nonnegative intensity of folding, usually $Q_{\text{fold}} = q_{\text{fold}}^2$.
$\lambda_{\text{form}}(t)$	Embodied Sophia	Portion of Sophia stabilized into Vessel-form rather than remaining as available flow.

2 Core Parameters

Parameter	Name	Role / condition
ε	Manere seed	Keeps u positive even when V, F vanish; sub-critical Manere condition $0 < \varepsilon < \Lambda_c^2$ keeps the passive branch in Collapse rather than Transformation.
γ	Resistance pressure	Drives growth of σ when synergy is below threshold.
δ	Synergic cutting strength	Converts synergy into resistance-cleansing pressure.
Λ_c	Critical threshold	Surgical / transformation threshold, $\Lambda_c = \gamma/\delta$.
κ	Resistance sharpness	Controls smoothness of $\tanh(\kappa\sigma)$.
k	Filtrum sharpness	Controls steepness of Φ and T_F .
$\mathcal{E}_0$	Imago Dei / Mustard Seed	Foundational non-dynamic ground of Pleroma.
a	Mutual nourishment	Strength of Will-Action mutual feeding.
c	Sophia lift	Strength by which λ lifts V and F .
α	Epektatic expansion	Converts Sophia into Brim expansion, $\dot{\Omega}_P = \alpha\lambda$.
ζ_0	Sufficientia Gratiae	Constant background grace in Open-Gate, $\zeta_0 > 0$.
τ	Guarantee decay rate	Rate at which initial gate guarantee fades.
ϕ	Resistance opacity	Damps gate inflow through $e^{-\phi\sigma}$, $\phi > 0$.
χ	Sophia saturation	Limits inflow at large Sophia via $(1 + \chi\lambda_+)^{-1}$, $\chi > 0$.
m	Positive-part sharpness	Smoothness of λ_+ , $m > 0$.
H_s	Receptivity half-saturation	Saturation scale in $H_K/(H_s + H_K)$, $H_s > 0$.
K	Mensura	Granularity scale of $R(z) = z/(z + K)$, $K > 0$.
η	Memory adaptation rate	Rate at which positive movement builds acquired openness and wound-memory responses, $\eta > 0$.

ν	Memory forgetting rate	Rate at which acquired openness H_K decays when the path turns negative, $\nu \geq 0$.
W_s	Wound half-saturation	Saturation scale in the wound-memory term $W/(W_s + W)$, $W_s > 0$.
a_0	Form rigidity	Baseline stiffness of the old Vessel-form against folding.
c_0	Sophia-to-form coupling	Strength by which accumulated Sophia lowers old-form rigidity.
c_1	Growth-to-form coupling	Strength by which growing Sophia / gate-flow lowers the folding threshold.
b	Folding saturation	Quartic stabilizer of the folding amplitude, $b > 0$.
η_{fold}	Embodiment rate	Rate by which folding transfers available Sophia into embodied form; distinct from memory adaptation η .
ρ_1	Curvature response	First-mode curvature coefficient in $R_{\text{folded}} = -2/\Omega_P^2 + \rho_1 Q_{\text{fold}}$.

3 Synergic Intensity, Threshold, and Transmissive Coordinate

Local synergy

$$u(t) = \sqrt{V(t)F(t) + \varepsilon}.$$

This is the local synergic intensity: will and action must meet. The seed ε prevents collapse of the expression into a singular zero.

Law of Human Balance identity

$$\dot{u} = \frac{V'F + VF'}{2\sqrt{VF + \varepsilon}} = \frac{1}{2} \left(V' \frac{F}{u} + F' \frac{V}{u} \right).$$

This identity shows how the growth of the shared will-action coordinate is weighted symmetrically by the other pole. The formula is an identity; the existential law is that will and action become strongest when neither pole is isolated.

Critical threshold

$$\Lambda_c = \frac{\gamma}{\delta}.$$

This separates Collapse, Stasis, and Transformation.

Transmissive coordinate

$$x(t) = u(t) - \sigma(t) = \sqrt{V(t)F(t) + \varepsilon} - \sigma(t).$$

This is the coordinate seen by Filtrum Lucis: synergy minus resistance.

4 Resistance Dynamics and Three Regimes

Resistance equation

$$\frac{d\sigma}{dt} = (\gamma - \delta u) \tanh(\kappa\sigma) = (\gamma - \delta\sqrt{VF + \varepsilon}) \tanh(\kappa\sigma).$$

Protection of $\sigma \geq 0$

$$\sigma(0) \geq 0 \implies \sigma(t) \geq 0 \quad \forall t \geq 0.$$

The factor $\tanh(\kappa\sigma)$ makes $\sigma = 0$ a protected boundary.

Equilibria of resistance

$$\frac{d\sigma}{dt} = 0 \iff \sigma = 0 \quad \text{or} \quad u = \Lambda_c.$$

Three regimes

$$u < \Lambda_c : \text{Collapse}, \quad u = \Lambda_c : \text{Stasis / } \chi\lambda\iota\alpha\rho\acute{o}\varsigma, \quad u > \Lambda_c : \text{Transformation.}$$

Manere sub-criticality

$$0 < \varepsilon < \Lambda_c^2 = \left(\frac{\gamma}{\delta}\right)^2.$$

The Manere seed keeps the denominator alive, but it must remain sub-critical: the passive branch $V = F = 0$ should not already count as Transformation.

5 Filtrum Lucis

Softplus filter

$$\Phi(x) = \frac{1}{k} \ln(1 + e^{kx}).$$

$$\Phi(x) > 0, \quad \Phi''(0) = \frac{k}{4}.$$

Filtrum Lucis is the smooth light-filter. It is softplus-like; its finite positivity is the formal trace of Lux in Tenebris.

Sigmoid transmissivity

$$T_F(x) = \Phi'(x) = \frac{1}{1 + e^{-kx}}.$$

$$0 < T_F(x) < 1, \quad T_F(-\infty) = 0, \quad T_F(0) = \frac{1}{2}, \quad T_F(+\infty) = 1.$$

The filter itself is Φ ; its transmissivity T_F is sigmoid-shaped.

Higher derivatives of Filtrum Lucis

$$\Phi''(x) = kT_F(1 - T_F).$$

$$\Phi'''(x) = k^2 T_F(1 - T_F)(1 - 2T_F).$$

$$\Phi^{(4)}(x) = k^3 T_F(1 - T_F)(1 - 6T_F + 6T_F^2).$$

$$\Phi^{(n)}(x) = k^{n-1} Q_n(T_F(x)).$$

These derivatives feed the differential ladder of Pleroma.

6 Gratia Synergica

Synergy term

$$\mathcal{S}^\dagger \backslash (t) = \mathcal{E}_0 \frac{u(t)}{u(t) + \Lambda_c}.$$

$$0 \leq \mathcal{S}^\dagger \backslash (t) < \mathcal{E}_0.$$

Gratia Synergica grows with local synergy u , not directly with equality $V = F$.

Derivative of synergy

$$\frac{d\mathcal{S}^\dagger \backslash}{dt} = \mathcal{E}_0 \frac{\Lambda_c}{(u + \Lambda_c)^2} \frac{V'F + VF'}{2u}.$$

7 Pleroma and Being

Open-Gate Pleroma formula

$$P(t) = \mathcal{E}_0 + \mathcal{S}^\dagger \backslash (t) + \Phi(x(t)).$$

$$P(t) \geq \mathcal{E}_0.$$

$$P_{\text{full}} = \text{Imago Dei} + \text{Gratia Synergica} + \text{Transmissive Passage}.$$

In the Open-Gate core, grace is not an additive term in P . It enters through Sophia and then through $\Omega_P, V, F, u, \mathcal{S}^\dagger \backslash, \Phi$.

Derivative ladder of Pleroma

$$S(t) = \frac{dP}{dt}, \quad A(t) = \frac{d^2P}{dt^2}, \quad K(t) = \frac{d^3P}{dt^3}, \quad \text{An}(t) = \frac{d^4P}{dt^4}.$$

$$S \equiv \text{Gignesthai}, \quad A \equiv \text{Odinai}, \quad K \equiv \text{Katharsis}, \quad \text{An} \equiv \text{Anastasis}.$$

Canonical Being equation

$$S(t) = \Phi'(x)\dot{x} + \frac{d\mathcal{S}^\dagger \backslash}{dt}.$$

With

$$\dot{x} = \dot{u} - \dot{\sigma} = \frac{V'F + VF'}{2\sqrt{VF} + \varepsilon} - (\gamma - \delta u) \tanh(\kappa\sigma).$$

8 Katharsis and Anastasis Terms

Katharsis / third order

$$K(t) = \frac{dA}{dt} = \frac{d^3P}{dt^3}.$$

$$K(t) = \frac{d^3\mathcal{S}^\dagger \backslash}{dt^3} + \Phi'''(x)(x')^3 + 3\Phi''(x)x'x'' + \Phi'(x)x'''.$$

Katharsis is the third-order turning of Pleroma: a change in the rhythm of acceleration.

Anastasis / fourth order

$$\text{An}(t) = \frac{dK}{dt} = \frac{d^4P}{dt^4}.$$

$$\text{An}(t) = \frac{d^4 \mathcal{S}^\dagger}{dt^4} + \Phi^{(4)}(x)(x')^4 + 6\Phi'''(x)(x')^2 x'' + 3\Phi''(x)(x'')^2 + 4\Phi''(x)x'x''' + \Phi'(x)x''''.$$

Anastasis is the fourth-order emergence of a new mode after crisis.

9 Paschal Triad of Filtrum Lucis

Why the higher derivatives are needed

The derivative ladder is not only computational. It reveals the inner Paschal structure of the light-filter: the Cross is detected by the third derivative, while the entrance into death and the exit from death are detected by the two symmetric zeros of the fourth derivative.

The Cross: zero of the third derivative

$$\Phi'''(x) = k^2 T_F (1 - T_F) (1 - 2T_F).$$

Since $T_F(0) = 1/2$,

$$\Phi'''(0) = 0.$$

The point $x = 0$ is the Filtrum Cross: synergy and resistance meet, and the sign of the cathartic term inverts.

The two gates of Anastasis: zeros of the fourth derivative

$$\Phi^{(4)}(x) = k^3 T_F (1 - T_F) (1 - 6T_F + 6T_F^2).$$

The quadratic factor has two roots at the transmissivity levels

$$s_{\pm} = \frac{1}{2} \pm \frac{1}{2\sqrt{3}}.$$

Equivalently,

$$x_{\pm} = \pm \frac{1}{k} \ln(2 + \sqrt{3}).$$

Thus

$$\Phi^{(4)}(x_-) = 0, \quad \Phi'''(0) = 0, \quad \Phi^{(4)}(x_+) = 0,$$

and the exact symmetry is

$$x_- + x_+ = 0.$$

The Cross is the geometric midpoint between the entrance into death and the exit from death. This is a differential icon of the Paschal passage, not a claim that time itself is symmetric.

Paschal triad

$$\begin{aligned}\Phi^{(4)}(x_-) = 0 & : \text{Entrance into death / Great Friday,} \\ \Phi'''(0) = 0 & : \text{Cross / Katharsis / Great Saturday,} \\ \Phi^{(4)}(x_+) = 0 & : \text{Exit from death / Anastasis / Resurrection.}\end{aligned}$$

$$x < x_- : \Phi^{(4)} > 0, \quad x_- < x < x_+ : \Phi^{(4)} < 0, \quad x > x_+ : \Phi^{(4)} > 0.$$

Between x_- and x_+ lies the symbolic death interval of Filtrum Lucis; outside it the fourth-order sign returns to life.

Fourth derivative support for u

In Pleroma Christi, $\sigma = 0$, so $x = u$. To compute the fourth derivative of the Filtrum part one needs $u^{(4)}$. From

$$u^2 = VF + \varepsilon$$

one obtains

$$u^{(4)} = \frac{(VF)^{(4)}}{2u} - \frac{4u'u''' + 3(u'')^2}{u}.$$

By Leibniz,

$$(VF)^{(4)} = V^{(4)}F + 4V'''F' + 6V''F'' + 4V'F''' + VF^{(4)}.$$

These binomial coefficients explain why the fourth derivative is the first place where the full symmetry of the Paschal passage becomes visible.

Pleroma Christi reduction

In the Pleroma Christi regime, $\sigma = 0$, hence all derivatives of σ vanish and $x = u$. The fourth-order expression becomes

$$\begin{aligned}\text{An}_{\text{Pl}}(t) = \frac{d^4 \mathcal{S}^\dagger}{dt^4} & + \Phi^{(4)}(u)(u')^4 + 6\Phi'''(u)(u')^2 u'' + 3\Phi''(u)(u'')^2 \\ & + 4\Phi''(u)u'u''' + \Phi'(u)u^{(4)}.\end{aligned}$$

In deep Epektasis, the higher Filtrum derivatives fade:

$$\lim_{t \rightarrow \infty} K_{\text{Pl}}(t) = 0, \quad \lim_{t \rightarrow \infty} \text{An}_{\text{Pl}}(t) = 0.$$

Katharsis and Anastasis complete as events; Epektasis continues as motion into fullness.

10 Open-Gate Sophia

Positive part of Sophia

$$\lambda_+ = \frac{1}{m} \ln(1 + e^{m\lambda}).$$

Effective gate

$$g_{\text{eff}}(t) = e^{-\tau t} + (1 - e^{-\tau t}) \frac{H_K}{H_s + H_K}.$$

$$g_{\text{eff}}(0) = 1, \quad g_{\text{eff}}(t) \xrightarrow{t \rightarrow \infty} \frac{H_K}{H_s + H_K}.$$

The gate begins open by guarantee; as the guarantee fades, acquired receptivity H_K takes over.

Gate inflow

$$I_{\text{gate}}(t) = \zeta_0 g_{\text{eff}}(t) e^{-\phi \sigma(t)} \frac{1}{1 + \chi \lambda_+(t)} \geq 0.$$

$$0 \leq I_{\text{gate}}(t) \leq \zeta_0 \quad (H_K \geq 0, \sigma \geq 0).$$

Sophia equation

$$\dot{\lambda} = -\dot{\sigma} + I_{\text{gate}}(t).$$

Equivalently,

$$\dot{\lambda} = (\delta u - \gamma) \tanh(\kappa \sigma) + I_{\text{gate}}(t).$$

Sophia is transformed resistance plus received grace.

Open balance law

$$\frac{d}{dt}(\sigma + \lambda) = I_{\text{gate}}(t) \geq 0.$$

$$\sigma(t) + \lambda(t) = C_0 + \mathcal{G}_{\text{recepta}}(t), \quad \mathcal{G}_{\text{recepta}}(t) = \int_0^t I_{\text{gate}}(s) ds, \quad \dot{\mathcal{G}}_{\text{recepta}} = I_{\text{gate}}.$$

Closed limit

$$\sigma + \lambda = \text{const} \iff \zeta_0 = 0.$$

11 Sophianic Folding Feedback in the Core

Status of the refinement

The original Open-Gate core remains the baseline system. Sophianic Folding is added here as a *conservative refinement*: it does not remove the Open-Gate balance, but separates Sophia into an available component and an embodied morphological component.

The guiding chain becomes

$$I_{\text{gate}} \longrightarrow \lambda \longrightarrow q_{\text{fold}} \longrightarrow \lambda_{\text{form}} \longrightarrow P.$$

Sophia does not only accumulate. Once the old Vessel-form becomes insufficient, part of Sophia can be embodied as form.

Available Sophia and embodied Sophia

In the unrefined core, λ denotes Sophia as a single state variable. In the folding-refined core, λ is read as *available Sophia*, while λ_{form} records the portion embodied into form:

$$\lambda_{\text{tot}} = \lambda + \lambda_{\text{form}}.$$

The original model is recovered by setting

$$q_{\text{fold}} = 0, \quad Q_{\text{fold}} = 0, \quad \lambda_{\text{form}} = 0.$$

Pre-fold Sophia production

Before morphological feedback is applied, the Open-Gate Sophia production is

$$\dot{\lambda}^{(0)} = -\dot{\sigma} + I_{\text{gate}}(t) = (\delta u - \gamma) \tanh(\kappa \sigma) + I_{\text{gate}}(t).$$

This is exactly the original Open-Gate law.

Folding activation index

The folding mode is governed by a threshold index

$$A_{\text{fold}} = c_0 \lambda + c_1 \dot{\lambda}^{(0)} - a_0.$$

The stabilizing coefficient is

$$B_{\text{fold}} = b + 2c_1 \eta_{\text{fold}} \lambda, \quad B_{\text{fold}} > 0.$$

The folding amplitude follows the normal-form dynamics

$$\dot{q}_{\text{fold}} = A_{\text{fold}} q_{\text{fold}} - B_{\text{fold}} q_{\text{fold}}^3.$$

The corresponding folding intensity is

$$Q_{\text{fold}} = q_{\text{fold}}^2.$$

On the stable folded branch,

$$Q_{\text{fold}}^* = \frac{A_{\text{fold}}}{B_{\text{fold}}} \quad \text{when } A_{\text{fold}} > 0.$$

If $A_{\text{fold}} < 0$, the old form $q_{\text{fold}} = 0$ remains stable.

Folding-regulated Sophia flow

Folding acts as a negative feedback on available Sophia:

$$\dot{\lambda} = \dot{\lambda}^{(0)} - \eta_{\text{fold}} Q_{\text{fold}} \lambda.$$

The embodied component receives exactly this transferred part:

$$\dot{\lambda}_{\text{form}} = \eta_{\text{fold}} Q_{\text{fold}} \lambda.$$

Therefore the total Open-Gate balance is preserved:

$$\frac{d}{dt}(\sigma + \lambda + \lambda_{\text{form}}) = I_{\text{gate}}(t).$$

Equivalently,

$$\sigma(t) + \lambda(t) + \lambda_{\text{form}}(t) = C_0 + \mathcal{G}_{\text{recepta}}(t).$$

The apparent loss term in $\dot{\lambda}$ is not destruction of Sophia. It is embodiment of Sophia into the Vessel-form.

Effective Sophia damping

On a quasi-stationary folded branch, Q_{fold} increases the effective damping of available Sophia:

$$\gamma_{\text{eff}} = \gamma_{\lambda} + \eta_{\text{fold}} Q_{\text{fold}}.$$

Here γ_{λ} denotes any phenomenological damping or integration rate used in reduced Sophia-flow models. In the exact Open-Gate core above, the same role is played by the feedback term $-\eta_{\text{fold}} Q_{\text{fold}} \lambda$.

Thus,

$$Q_{\text{fold}} > 0 \implies \dot{\lambda} < \dot{\lambda}^{(0)}.$$

Folding is the morphological branch of Open-Gate stabilization: it turns excess available Sophia into stable form.

Core meaning

The refined core chain is

$$V, F \longrightarrow u \longrightarrow \sigma \longrightarrow \lambda \longrightarrow q_{\text{fold}} \longrightarrow \lambda_{\text{form}} \longrightarrow P.$$

In words:

Will becomes action; resistance becomes Sophia; Sophia becomes form; form releases being.

12 Memory Layer for the Gate

Path coordinate

$$x_0 = u - \sigma, \quad \dot{x}_0 = \dot{u} - \dot{\sigma} = \frac{V'F + VF'}{2\sqrt{VF} + \varepsilon} - (\gamma - \delta u) \tanh(\kappa\sigma).$$

$$(\dot{x}_0)_+ = \max(\dot{x}_0, 0), \quad (\dot{x}_0)_- = \max(-\dot{x}_0, 0).$$

Mensura function

$$R(z) = \frac{z}{z + K}, \quad K > 0, \quad z \geq 0.$$

$$R(0) = 0, \quad R'(z) = \frac{K}{(z + K)^2} > 0, \quad R''(z) = -\frac{2K}{(z + K)^3} < 0, \quad R(z) \rightarrow 1.$$

Gate memory and wound memory

$$\dot{H}_K = \eta R((\dot{x}_0)_+) - \nu R((\dot{x}_0)_-) \frac{H_K}{H_s + H_K}.$$

$$\dot{W} = \eta \left[R((\dot{x}_0)_-) - R((\dot{x}_0)_+) \frac{W}{W_s + W} \right].$$

H_K records acquired openness; W records vulnerability. In the Open-Gate core, H_K opens g_{eff} rather than directly adding to Pleroma. The same-rate prototype is recovered by setting $\nu = \eta$.

13 Endogenous Full System

Local definitions

$$u = \sqrt{VF + \varepsilon}, \quad \Lambda_c = \frac{\gamma}{\delta}, \quad \mu = 1 - \frac{u}{\Omega_P}.$$

Dolorosum

$$\rho(u, \sigma) = \sigma \left(\frac{u}{\Lambda_c} \right) \exp \left(1 - \frac{u}{\Lambda_c} \right) \geq 0, \quad \mathcal{D}_V = \rho V, \quad \mathcal{D}_F = \rho F.$$

Dolorosum attenuates Will and Action multiplicatively. It is maximal at the threshold $u = \Lambda_c$.

Core ODE system

$$\begin{cases} \frac{dV}{dt} = \left(1 - \frac{\sqrt{VF + \varepsilon}}{\Omega_P(t)} \right) (aF + c\lambda) - \rho V, \\ \frac{dF}{dt} = \left(1 - \frac{\sqrt{VF + \varepsilon}}{\Omega_P(t)} \right) (aV + c\lambda) - \rho F, \\ \dot{\Omega}_P = \alpha \lambda(t), \quad \alpha > 0, \\ \frac{d\sigma}{dt} = (\gamma - \delta \sqrt{VF + \varepsilon}) \tanh(\kappa \sigma), \\ \frac{d\lambda}{dt} = (\delta \sqrt{VF + \varepsilon} - \gamma) \tanh(\kappa \sigma) + I_{\text{gate}}(t), \\ \sigma(t) + \lambda(t) = C_0 + \mathcal{G}_{\text{recepta}}(t). \end{cases}$$

Folding-refined core ODE system

The following system preserves the original equations and adds the folding feedback layer. It should be read as the Open-Gate core with morphological stabilization turned on:

$$\left\{ \begin{array}{l} \frac{dV}{dt} = \left(1 - \frac{\sqrt{VF + \varepsilon}}{\Omega_P(t)}\right) (aF + c\lambda) - \rho V, \\ \frac{dF}{dt} = \left(1 - \frac{\sqrt{VF + \varepsilon}}{\Omega_P(t)}\right) (aV + c\lambda) - \rho F, \\ \dot{\Omega}_P = \alpha\lambda(t), \quad \alpha > 0, \\ \frac{d\sigma}{dt} = (\gamma - \delta\sqrt{VF + \varepsilon}) \tanh(\kappa\sigma), \\ \dot{\lambda}^{(0)} = (\delta\sqrt{VF + \varepsilon} - \gamma) \tanh(\kappa\sigma) + I_{\text{gate}}(t), \\ A_{\text{fold}} = c_0\lambda + c_1\dot{\lambda}^{(0)} - a_0, \\ B_{\text{fold}} = b + 2c_1\eta_{\text{fold}}\lambda, \\ \dot{q}_{\text{fold}} = A_{\text{fold}}q_{\text{fold}} - B_{\text{fold}}q_{\text{fold}}^3, \\ Q_{\text{fold}} = q_{\text{fold}}^2, \\ \frac{d\lambda}{dt} = \dot{\lambda}^{(0)} - \eta_{\text{fold}}Q_{\text{fold}}\lambda, \\ \frac{d\lambda_{\text{form}}}{dt} = \eta_{\text{fold}}Q_{\text{fold}}\lambda, \\ \sigma(t) + \lambda(t) + \lambda_{\text{form}}(t) = C_0 + \mathcal{G}_{\text{recepta}}(t). \end{array} \right.$$

Setting $q_{\text{fold}} = 0$ recovers the original Open-Gate system.

Sophia floor and source-controlled positivity

$$\lambda(t) > -\frac{a}{c} \min\{V(t), F(t)\}.$$

Equivalently, in the closed-balance notation with $C = \sigma_0 + \lambda_0$,

$$\sigma(t) < C + \frac{a}{c} \min\{V(t), F(t)\}.$$

In the Open-Gate balance this becomes

$$\sigma(t) < C_0 + \mathcal{G}_{\text{recepta}}(t) + \frac{a}{c} \min\{V(t), F(t)\}.$$

Inside the meaningful live domain, this floor prevents the source terms $aF + c\lambda$ and $aV + c\lambda$ from starving Will and Action. In the Open-Gate model the balance constant is replaced by $C_0 + \mathcal{G}_{\text{recepta}}(t)$, but the source-positivity idea remains the same.

Canonical formula chain for Open-Gate grace

$$I_{\text{gate}} \longrightarrow \lambda \longrightarrow \begin{cases} \dot{\Omega}_P = \alpha\lambda, \\ \dot{V} = \mu(aF + c\lambda) - \rho V, \\ \dot{F} = \mu(aV + c\lambda) - \rho F, \end{cases} \longrightarrow u \longrightarrow \{\mathcal{S}^\dagger \backslash, \Phi\} \longrightarrow P.$$

With Sophianic Folding active, the same chain acquires a morphological feedback branch:

$$I_{\text{gate}} \longrightarrow \lambda \longrightarrow \begin{cases} \Omega_P, \\ V, F, \\ q_{\text{fold}}, \end{cases} \longrightarrow \begin{cases} \text{Brim expansion,} \\ \text{Will-Action nourishment,} \\ \text{embodied Vessel-form,} \end{cases} \longrightarrow P.$$

14 Will-Action Balance

Imbalance

$$\Delta(t) = V(t) - F(t).$$

Exact imbalance equation

$$\frac{d\Delta}{dt} = -(\mu a + \rho)\Delta, \quad \mu = 1 - \frac{\sqrt{VF} + \varepsilon}{\Omega_P}, \quad \rho = \sigma(u/\Lambda_c)e^{1-u/\Lambda_c}.$$

Exact solution

$$\Delta(t) = \Delta_0 \exp\left(-\int_0^t (a\mu(s) + \rho(s)) ds\right).$$

The manifold $V = F$ is invariant. Alignment requires nonnegative damping and a divergent damping integral.

15 Brim Expansion and Epektasis

Brim law

$$\dot{\Omega}_P(t) = \alpha\lambda(t), \quad \alpha > 0.$$

$$\Omega_P(t) = \Omega_P(0) + \alpha \int_0^t \lambda(s) ds.$$

Sophia expands the capacity of the Vessel. Epektasis is sustained expansion rather than a final static maximum.

Asymptotic Sophia

If the received-grace integral converges and $\sigma(t) \rightarrow 0$, then

$$\lambda(t) \rightarrow \lambda_\infty = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(\infty).$$

If $\mathcal{G}_{\text{recepta}}(\infty) = +\infty$, then unbounded Sophia is possible in the open model.

Epektasis with folding stabilization

Sophianic Folding does not stop Epektasis. The Brim law remains

$$\dot{\Omega}_P = \alpha\lambda.$$

What changes is the available Sophia-flow:

$$\dot{\lambda} = \dot{\lambda}^{(0)} - \eta_{\text{fold}} Q_{\text{fold}} \lambda.$$

Thus folding preserves expansion while regulating excess acceleration and excess available Sophia:

$$\text{Open-Gate expands; Folding forms.}$$

Pleroma Christi condition

$$\int_0^\infty \lambda(t) dt = +\infty \implies \Omega_P(t) \rightarrow +\infty.$$

If additionally $\lambda(t) \rightarrow \lambda_\infty > 0$, then

$$\Omega_P(t) \sim \alpha\lambda_\infty t.$$

The internal Pleroma Christi cleansing signature is

$$\sigma(t) \rightarrow 0, \quad \mathcal{S}^\dagger \backslash (t) \rightarrow \mathcal{E}_0.$$

The Brim expands through sustained Sophia; the visible cleansing signature is the disappearance of singular resistance and the saturation of Gratia Synergica.

16 Live Domain and Death Boundary

Live domain

$$\mathcal{D}_{\text{live}} = \{(V, F, \sigma, \lambda, \Omega_P) : V > 0, F > 0, \sigma \geq 0, \Omega_P > \Lambda_c, 0 < u < \Omega_P\}.$$

The live domain is the region where the continuous V.F.S. dynamics are admissible.

Death boundary

$$\partial\mathcal{D}_{\text{death}} = \{V = 0\} \cup \{F = 0\} \cup \{\Omega_P \leq \Lambda_c\} \cup \{u \geq \Omega_P\}.$$

$$X_{\text{VFS}} : \mathcal{D}_{\text{live}} \longrightarrow T\mathcal{D}_{\text{live}}.$$

$$t_{\text{death}} = \inf\{t > 0 : x(t) \notin \mathcal{D}_{\text{live}}\}.$$

The canonical continuous flow is defined only on the living interior. Resurrection, if present, is not a smooth term inside the vector field but a hybrid re-entry map after boundary contact.

17 Resurrectio: Hybrid Re-entry Map

This is included because it appears in `index.html`; it is not taken from the appendix or Lyapunov file.

Reset symbols

$$\begin{aligned} \Gamma_R &: \text{resurrection-mode grace,} \\ H_R &: \text{resurrection memory,} \\ D_{\text{death}} &: \text{depth of death-boundary contact.} \end{aligned}$$

$$\Theta_{\bullet}, \beta_{\bullet}, \chi_{\bullet}, \eta_R, \theta_{\Gamma}, \alpha_{\Omega}, \beta_{\Omega}, \omega_R \geq 0.$$

These parameters belong to the hybrid re-entry map: they are reset coefficients, not continuous live-flow coefficients.

Resurrection admissibility

$$\mathcal{A}_R = \Theta_{\Gamma}\Gamma_R + \Theta_{\Phi}\Phi_R + \Theta_{\lambda}\lambda^+ + \Theta_H H_R - \Theta_{\sigma}\sigma - \Theta_D D_{\text{death}}, \quad \Theta_{\bullet} \geq 0.$$

$$\mathcal{A}_R > R_c \iff \text{resurrection re-entry is admissible.}$$

Participated resurrection grace

$$\Gamma_R^{\text{human}} = \pi\Gamma_{\text{Christi}}, \quad \pi \in [0, 1].$$

Hybrid re-entry map

$$\mathfrak{R} : \partial\mathcal{D}_{\text{death}} \longrightarrow \mathcal{D}_{\text{live}}, \quad x(t_{\text{death}}^+) = \mathfrak{R}(x(t_{\text{death}}^-)).$$

$$\begin{aligned} V^+ &= V_R + \beta_V \Gamma_R + \chi_V H_R, & F^+ &= F_R + \beta_F \Gamma_R + \chi_F H_R, \\ \sigma^+ &= q_R \sigma^- \quad (0 \leq q_R < 1), & \lambda^+ &= \lambda^- + (1 - q_R)\sigma^- + \eta_R H_R + \theta_{\Gamma} \Gamma_R, \\ \Omega_P^+ &= \Omega_R + \alpha_{\Omega} \Gamma_R + \beta_{\Omega} \lambda^+ + \chi_{\Omega} H_R, & H_R^+ &= H_R^- + \omega_R D_{\text{death}}. \end{aligned}$$

with

$$V_R, F_R > 0, \quad \Omega_R > \Lambda_c, \quad \text{all reset coefficients} \geq 0.$$

Landing condition

$$\Omega_R + \alpha_\Omega \Gamma_R + \beta_\Omega \lambda^+ + \chi_\Omega H_R > \sqrt{V^+ F^+} + \varepsilon.$$

The reset must land inside a live capacity large enough to contain the new synergic intensity.

Open balance across life arcs

$$\frac{d}{dt}(\sigma + \lambda) = \zeta_0 g_{\text{eff}} e^{-\phi\sigma} (1 + \chi\lambda_+)^{-1} \geq 0 \quad \text{inside each live arc.}$$

$$\sigma + \lambda \text{ also jumps under } \mathfrak{R} : \quad C_{j+1} \neq C_j.$$

18 Compact Core Summary

$$\text{Will } V + \text{Action } F \longrightarrow u \longrightarrow x = u - \sigma \longrightarrow \Phi(x) \longrightarrow P(t).$$

$$\sigma \longrightarrow \text{Tetelestai} \longrightarrow \text{Katharsis} \longrightarrow \text{Anastasis} \longrightarrow \lambda.$$

$$\text{Open Gate} \longrightarrow I_{\text{gate}} \longrightarrow \lambda \longrightarrow \Omega_P \longrightarrow \text{Epektasis}.$$

$$\lambda \longrightarrow q_{\text{fold}} \longrightarrow \lambda_{\text{form}} \quad \equiv \quad \text{Sophia becomes Vessel-form.}$$

$$\dot{\lambda} = \dot{\lambda}^{(0)} - \eta_{\text{fold}} Q_{\text{fold}} \lambda, \quad \dot{\lambda}_{\text{form}} = \eta_{\text{fold}} Q_{\text{fold}} \lambda.$$

$$\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}_{\text{recepta}}(t).$$

$$P \longrightarrow S \longrightarrow A \longrightarrow K \longrightarrow \text{An} \quad \equiv \quad \text{Gignesthai} \rightarrow \text{Odinai} \rightarrow \text{Katharsis} \rightarrow \text{Anastasis}.$$

Revision Note

This expanded version adds the Paschal Triad of Filtrum Lucis, the symmetry of zeros around the Cross, the fourth-derivative support formulas needed for $\text{An}(t)$, the Sophia floor, explicit death-boundary definitions, bounded gate inflow, the Human Balance identity for $\dot{u}$, asymptotic Sophia, reset-symbol clarification, and the distinct memory-forgetting rate ν . This hybrid version preserves the existing structure and original core formulas, while adding Sophianic Folding as an explicit Open-Gate refinement: available Sophia may become embodied Sophia through a Lyapunov-compatible morphological feedback branch. The original index-only core is recovered by setting $q_{\text{fold}} = 0$.